

Black-Scholes Pricing Memo

ETERNAL (Zomato) — NSE Option Chain Analysis

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Contents

1	The Underlying	2
2	Historical Volatility Estimation	2
3	Risk-Free Rate	3
4	Black-Scholes Pricing	3
4.1	Input Parameters	3
4.2	Intermediate Values	3
4.3	Theoretical Prices	4
5	Comparison with Market Prices	4
6	Option Greeks	4
6.1	Computed Greeks for ETERNAL Call	4
6.2	Price Prediction After Hypothetical Move	4
7	American-Style Adjustment	5
8	Reflection	5
A	Python Code	6

1 The Underlying

We analyse **ETERNAL** (NSE: ETERNAL), the parent company of Zomato, a high-growth consumer technology firm listed on the NSE.

- Spot Price $S = \text{INR } 256.75$ (as on 5-Jun-2026)
- Strike Price $K = \text{INR } 255$ (near ATM)
- Expiry Date = 30-Jun-2026
- Time to Expiry $T = \frac{22}{365} = 0.0603$ years

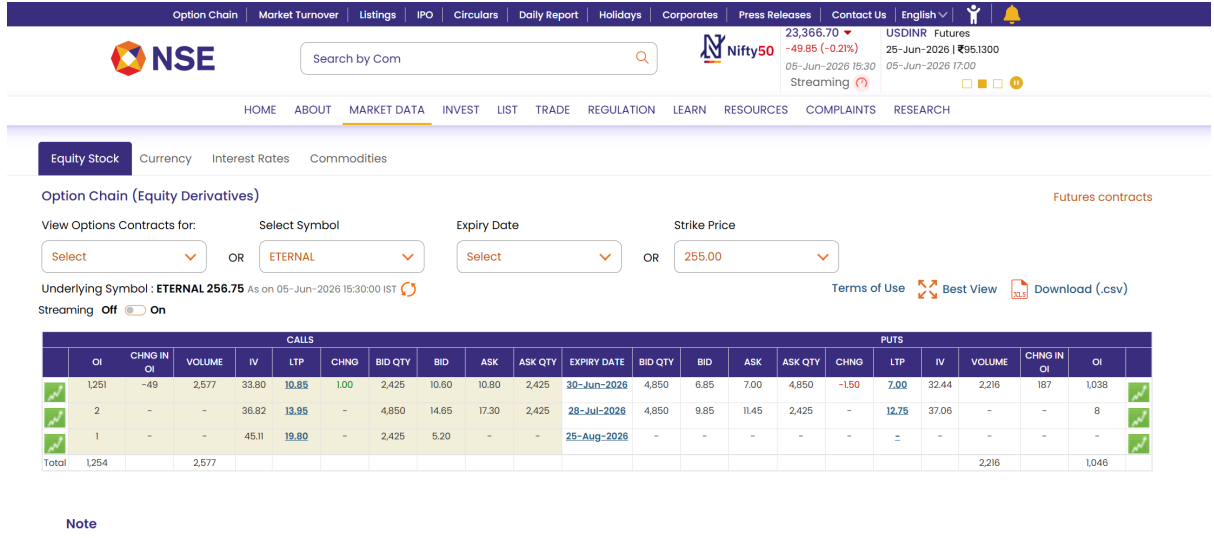


Figure 1: ETERNAL Option Chain from NSE India (5-Jun-2026)

2 Historical Volatility Estimation

We compute 30-day historical volatility using 22 trading days of ETERNAL closing prices sourced from NSE India's historical data.

Daily log returns are calculated as:

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right) \quad (1)$$

Annualized volatility is then:

$$\sigma = \text{std} r_t \times \sqrt{252} \quad (2)$$

- Daily $\sigma = 0.020492$
- Annualized $\sigma = 32.53\%$
- Market Implied Volatility (Call) = 33.80%
- Market Implied Volatility (Put) = 32.44%

The historical and implied volatilities are closely aligned, suggesting the market does not anticipate any sharp deviation from recent price behaviour in ETERNAL. The small premium in call IV (33.80% vs 32.53%) may reflect a modest volatility risk premium charged by option writers.

3 Risk-Free Rate

We use the 91-day Government of India T-bill yield as the risk-free rate, as it is the most liquid and widely quoted short-term government instrument in India. Unlike bank deposits, T-bills carry no default risk as they are backed by the Government of India.

- 91-day T-bill yield = 5.56% per annum
- Continuously compounded rate:

$$r = \ln(1 + 0.0556) = \mathbf{5.41\%} \quad (3)$$

The continuously compounded form is required by the Black-Scholes formula, as the model assumes continuous trading and continuous compounding of returns.

4 Black-Scholes Pricing

The Black-Scholes formula for a European call and put is:

$$C = S \cdot N(d_1) - K e^{-rT} \cdot N(d_2) \quad (4)$$

$$P = K e^{-rT} \cdot N(-d_2) - S \cdot N(-d_1) \quad (5)$$

Where:

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2) \cdot T}{\sigma\sqrt{T}} \quad (6)$$

$$d_2 = d_1 - \sigma\sqrt{T} \quad (7)$$

4.1 Input Parameters

- $S = 256.75$, $K = 255$, $T = 0.0603$
- $r = 0.0541$, $\sigma = 0.3253$

4.2 Intermediate Values

- $d_1 = 0.1664$
- $d_2 = 0.0865$
- $N(d_1) = 0.5661$
- $N(d_2) = 0.5345$

4.3 Theoretical Prices

- **Call Price** = Rs. 9.49
- **Put Price** = Rs. 6.91

5 Comparison with Market Prices

Option	BS Theoretical	Market LTP	Deviation
Call	Rs. 9.49	Rs. 10.85	+Rs. 1.36 (+14.3%)
Put	Rs. 6.91	Rs. 7.00	+Rs. 0.09 (+1.3%)

Table 1: Black-Scholes Theoretical vs Market Prices for ETERNAL

The call option is trading at a significant premium of 14.3% above its theoretical price. This deviation can be attributed to two reasons. First, the market may hold a bullish view on ETERNAL, expecting the stock to close well above Rs. 255 with high probability. Second, the market implied volatility (33.80%) is slightly higher than historical volatility (32.53%), suggesting option writers are charging a modest volatility risk premium. The put deviation is negligible at 1.3%, suggesting the market is fairly pricing downside risk.

6 Option Greeks

6.1 Computed Greeks for ETERNAL Call

Greek	Symbol	Value	Interpretation
Delta	Δ	0.5661	Call gains Rs. 0.57 per Rs. 1 rise in S
Gamma	Γ	0.0192	Δ increases by 0.0192 per Rs. 1 rise in S
Vega	ν	0.2480	Call gains Rs. 0.25 per 1% rise in σ
Theta	Θ	-0.2035	Call loses Rs. 0.20 per day from time decay

Table 2: Option Greeks for ETERNAL Call (K=255, T=22 days)

6.2 Price Prediction After Hypothetical Move

We predict the new call price after a 1% rise in S and 1% rise in σ using the Greek approximation, analogous to Newton's equation of motion where Δ acts as velocity and Γ as acceleration:

$$\Delta C \approx \Delta \cdot \Delta S + \frac{1}{2} \cdot \Gamma \cdot \Delta S^2 + \nu \cdot \Delta \sigma \quad (8)$$

- $\Delta S = 1\% \times 256.75 = \text{Rs. } 2.5675$
- Price change from Δ and $\Gamma = \text{Rs. } 1.5167$

- Price change from Vega = Rs. 0.2480
- **Predicted Call Price** = $9.49 + 1.5167 + 0.2480 = \text{Rs. } 11.26$

7 American-Style Adjustment

While most single-stock NSE options are American-style, ETERNAL options are in fact **European-style**, meaning they can only be exercised at expiry. This makes Black-Scholes an **exact fit** for pricing rather than an approximation, as the model was originally derived for European options.

For completeness, we note that American-style single-stock options carry an early exercise premium when the underlying pays dividends. The holder may exercise just before the ex-dividend date to capture the dividend, since the option itself does not benefit from dividend payments. Since ETERNAL does not currently pay dividends, even if these were American-style options, the early exercise premium would be negligible and Black-Scholes would remain a good approximation.

8 Reflection

Applying Black-Scholes to a real option chain revealed several insights that formulas alone could not convey. First, I discovered that Delta is simply $N(d_1)$ something that is easy to miss when studying the formula in isolation but becomes obvious when you compute it step by step. Second, the Greek approximation for price prediction bears a striking resemblance to Newton's kinematic equation $s = ut + \frac{1}{2}at^2$, where Delta acts as velocity and Gamma as acceleration suggesting that the mathematics of motion and options pricing share a deep structural similarity. Third, applying the model to real NSE data revealed that historical and implied volatility can diverge, and that this gap itself carries information about market sentiment. Finally, the exercise highlighted that Black-Scholes is not a perfect pricing machine the 14.3% deviation in the call price reminds us that markets incorporate human sentiment, liquidity, and expectations that no closed-form model can fully capture.

References

1. Zerodha Varsity, *Options Theory for Professional Trading*, available at <https://zerodha.com/varsity/module/option-theory/>
2. NSE India, *Equity Derivatives Option Chain — ETERNAL*, available at <https://www.nseindia.com>, accessed 8-Jun-2026.
3. Black, F. and Scholes, M. (1973), *The Pricing of Options and Corporate Liabilities*, Journal of Political Economy, 81(3), 637–654.
4. Investopedia, *Black-Scholes Model*, available at <https://www.investopedia.com/terms/b/blackscholes.asp>

A Python Code

```
1 import pandas as pd
2 import numpy as np
3
4 df = pd.read_csv(["eternal.csv", encoding='utf-8', encoding_errors='ignore'])
5 df.columns = df.columns.str.strip()
6 df['Date'] = pd.to_datetime(df['Date'])
7 df = df.sort_values('Date')
8
9 df['Log Return'] = np.log(df['Close Price'] / df['Close Price'].shift(1))
10
11 recent = df.tail(22)
12 sigma_daily = recent['Log Return'].std()
13 sigma_annual = sigma_daily * np.sqrt(252)
14
15 print(f"Daily Sigma: {sigma_daily:.6f}")
16 print(f"Annualised Sigma (Historical Vol): {sigma_annual:.4f}")
```

Figure 2: Historical Volatility Computation (Python)

Daily Sigma: 0.020492
Annualised Sigma (Historical Vol): 0.3253

Figure 3: Historical Volatility Output

```

1  import numpy as np
2  from scipy.stats import norm
3
4  S = 256.75
5  K = 255
6  r = 0.0541
7  sigma = 0.3253
8  T = 22/365
9
10 d1 = (np.log(S/K) + (r + sigma**2/2) * T) / (sigma * np.sqrt(T))
11 d2 = d1 - sigma * np.sqrt(T)
12
13 call = S * norm.cdf(d1) - K * np.exp(-r*T) * norm.cdf(d2)
14 put = K * np.exp(-r*T) * norm.cdf(-d2) - S * norm.cdf(-d1)
15
16 print(f"d1: {d1:.4f}")
17 print(f"d2: {d2:.4f}")
18 print(f"N(d1): {norm.cdf(d1):.4f}")
19 print(f"N(d2): {norm.cdf(d2):.4f}")
20 print(f"BS Call Price: {call:.2f}")
21 print(f"BS Put Price: {put:.2f}")

```

Figure 4: Black-Scholes Pricing (Python)

```

d1: 0.1664
d2: 0.0865
N(d1): 0.5661
N(d2): 0.5345
BS Call Price: 9.49
BS Put Price: 6.91

```

Figure 5: Black-Scholes Output

```

1  from scipy.stats import norm
2  import numpy as np
3
4  S = 256.75
5  K = 255
6  r = 0.0541
7  sigma = 0.3253
8  T = 22/365
9
10 d1 = (np.log(S/K) + (r + sigma**2/2) * T) / (sigma * np.sqrt(T))
11 d2 = d1 - sigma * np.sqrt(T)
12
13 delta = norm.cdf(d1)
14 gamma = norm.pdf(d1) / (S * sigma * np.sqrt(T))
15 vega = S * norm.pdf(d1) * np.sqrt(T) / 100 # per 1% move in vol
16 theta = (-S * norm.pdf(d1) * sigma / (2 * np.sqrt(T)) - r * K * np.exp(-r*T) * norm.cdf(d2))
17
18 print(f"Delta: {delta:.4f}")
19 print(f"Gamma: {gamma:.6f}")
20 print(f"Vega: {vega:.4f}")
21 print(f"Theta: {theta:.4f}")

```

Figure 6: Greeks Computation (Python)

```

Delta: 0.5661
Gamma: 0.019188
Vega: 0.2480
Theta: -0.2035

```

Figure 7: Greeks Output